

SQL & DATABASES · DAY 3 OF 10

# Adding, changing and deleting data

INSERT, UPDATE, DELETE and saving safely

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Wednesday 7 October 2026

# Today

- After today you can:
  - Add rows with INSERT
  - Change rows with UPDATE
  - Remove rows with DELETE
  - Understand why WHERE is extra important with UPDATE and DELETE
  - Understand transactions: save (commit) or undo (rollback)

# Today's schedule

| Time        | What we do              |
|-------------|-------------------------|
| 09:00-09:10 | Warm-up                 |
| 09:10-09:50 | Theory (slides)         |
| 09:50-10:00 | Break                   |
| 10:00-10:55 | Code-along, part 1      |
| 10:55-11:05 | Break                   |
| 11:05-12:00 | Code-along, part 2      |
| 12:00-13:00 | Lunch                   |
| 13:00-14:30 | Lab, part 1             |
| 14:30-14:45 | Break                   |
| 14:45-16:15 | Lab, part 2             |
| 16:15-17:00 | Walkthrough and wrap-up |

# Four things you can do with data

- Developers call this CRUD: Create, Read, Update, Delete

| Command | Does   | Like...                          |
|---------|--------|----------------------------------|
| SELECT  | Read   | Looking in the cupboard          |
| INSERT  | Add    | Putting something in             |
| UPDATE  | Change | Fixing something that's in there |
| DELETE  | Remove | Throwing something out           |

## INSERT: add a new row

- First the column names, then the values, in the same order
- Columns you skip become NULL (or their DEFAULT)

```
INSERT INTO customers (customer_id, first_name, last_name, city)
VALUES (11, 'Noah', 'Svensson', 'Uppsala');
```

## Many rows at once

- One INSERT, many rows: commas between the brackets

```
INSERT INTO products (product_id, name, category, price, stock)
VALUES (13, 'Scarf', 'Accessories', 229, 15),
       (14, 'Gloves', 'Accessories', 199, 20);
```

# UPDATE: change existing rows

- SET: what should the new value be?
- WHERE: which rows should change?

```
UPDATE products  
SET price = 549  
WHERE product_id = 1;
```

# The most dangerous mistake in SQL

```
UPDATE products SET price = 0;
```

```
DELETE FROM customers;
```

**No WHERE = EVERY row is changed!**

# The safe way: SELECT first

```
-- 1. Check which rows you will hit  
SELECT * FROM products WHERE stock = 0;  
  
-- 2. Then change exactly those rows  
UPDATE products SET stock = 10 WHERE stock = 0;
```

## DELETE: remove rows

- Removes whole rows, not single values
- To empty one value, use UPDATE ... SET column = NULL
- A foreign key can stop you from deleting (good!)

```
DELETE FROM customers  
WHERE customer_id = 11;
```

# Transactions: save or undo

- A transaction is a group of changes that belong together
- COMMIT = save it all. ROLLBACK = undo it all
- Example: moving money. Take from one account AND add to another
- Either both happen, or none. Never just half!

# Transactions in DB Browser

- Your changes are not saved until you press Write Changes
- Made a mess? Revert Changes takes you back to the last save

| Button         | Shortcut | Means                          |
|----------------|----------|--------------------------------|
| Write Changes  | Ctrl+S   | COMMIT: save to the file       |
| Revert Changes | -        | ROLLBACK: undo since last save |

# Remember

- INSERT adds, UPDATE changes, DELETE removes
- Always use WHERE with UPDATE and DELETE
- Run a SELECT with the same WHERE first
- Ctrl+S = COMMIT (save). Revert Changes = ROLLBACK (undo)

## Your turn (1/2)

1. Add yourself as customer number 11 (use a made-up email).
2. Add two new products in one INSERT: Scarf (Accessories, 229 kr, 15 in stock) and Gloves (Accessories, 199 kr, 20 in stock).
3. Customer 7 (Emma) orders 2 Beanies (product 10, 179 kr). Make the order (order 16) and the order item.
4. Try to add an order item with quantity 0. Which rule stops you?
5. Order 12 has been shipped. Change its status.

## Your turn (2/2)

6. The Water Bottle (product 5) is back in stock: 50 pieces.
7. Raise the price of all Accessories by 10%.
8. Delete the cancelled order. Watch out: its items must go first! Why?
9. Finally: click Revert Changes so your data matches the teacher's again. Check: 15 orders?

## Coming up tomorrow

**Tomorrow we take a step back and learn how to design a good database from scratch.**